Homework 2

STAT 4714

Due: September 26, 2025

Question 1

(14pts) Let T denote the number of years until an electronic component fails. Given that the cumulative distribution function of T is

$$F_T(t) = \begin{cases} 0, & t < 1, \\ \frac{1}{4}, & 1 \leq t < 3, \\ \frac{1}{2}, & 3 \leq t < 5, \\ \frac{3}{4}, & 5 \leq t < 7, \\ 1, & t \geq 7, \end{cases}$$

find

1. (3pts) $P(T = 5)$;
2. (3pts) $P(T > 3)$;
3. (3pts) $P(1.4 < T < 6)$;
4. (5pts) $P(T \leq 5 \mid T \geq 2)$.

[*Hint:* The CDF $F_T(t)$ suggests that $F_T(2) = F_T(1.5) = F_T(1)$. It's recommended that you write the desired probability statements using the CDF. For the first part of this question, consider expressing $P(T = 5) = F_T(5) - F_T(4)$.]

Solution 1

1. As the hint suggests, we express $P(T = 5) = F_T(5) - F_T(4) = 3/4 - 1/2 = 1/4$. The reason we do this is because the CDF gives us the cumulative probability: $F_T(5) = P(T \leq 5)$. So, we need to remove the probabilities associated with the previous outcomes $(T = 1, T = 2, T = 3, T = 4)$ by subtracting $F_T(4) = P(T \leq 4)$.
2. Generally, it will be easier to solve probabilities containing inequalities if we structure it such that the random variable is *less than* a constant; we do this because then we can easily use the CDF. So, instead of solving

for $P(T > 3)$ directly, we express it as $P(T > 3) = 1 - P(T \leq 3) = 1 - F_T(3) = 1 - 1/2 = 1/2$.

3. Recall the result in class where $P(a < T < b) = F_T(b) - F_T(a)$. Thankfully, we are given the CDF: $P(1.4 < T < 6) = F_T(6) - F_T(1.4) = 3/4 - 1/4 = 1/2$.

4. This part is a bit more involved but nothing we can't handle! Ultimately, we want to write this complicated conditional probability with expressions containing the CDF. Note that

$$P(T \leq 5 \mid T \geq 2) = \frac{P(2 \leq T \leq 5)}{P(T \geq 2)}.$$

From here, we can use the tricks covered in the second and third part of this problem:

$$\frac{P(2 \leq T \leq 5)}{P(T \geq 2)} = \frac{P(T \leq 5) - P(T < 2)}{1 - P(T < 2)} = \frac{F_T(5) - F_T(1)}{1 - F_T(1)} = 2/3.$$

Note that we want to keep the value of $T = 2$ in the numerator and denominator. Therefore, we use $P(T < 2) = P(T \leq 1) = F_T(1)$.

Question 2

(11pts) Measurements of scientific systems are always subject to variation, some more than others. There are many structures for measurement error, and statisticians spend a great deal of time modeling these errors. Suppose the measurement error X of a certain physical quantity is decided by the density function

$$f(x) = \begin{cases} k(3 - x^2), & -1 \leq x \leq 1, \\ 0, & \text{elsewhere.} \end{cases}$$

1. (3pts) Determine k that renders $f(x)$ a valid density function.

2. (3pts) Find the probability that a random error in measurement is less than $1/2$.

3. (5pts) For this particular measurement, it is undesirable if the magnitude of the error (i.e., $|x|$) exceeds 0.8. What is the probability that this occurs?

Solution 2

1. We need to ensure that the PDF integrates to 1; let's use this fact to solve for k:

$$k \int_{-1}^{1} (3 - x^2) dx = k[3x - x^3/3]_{-1}^{1} = 16k/3 \overset{\text{set}}{=} 1,$$

which gives $k = 3/16$.

2. The question asks us to solve $P(X < 1/2) = F_X(1/2)$. We could solve this directly by integrating the PDF over the domain $[-1, 1/2]$ *or* we could solve using the CDF. Both options work. I'll show the latter just to get used to switching between the PDF and CDF:

$$F_X(x) = \int_{-1}^{x} \frac{3}{16}(3 - t^2)dt = \frac{1}{2} + \frac{9}{16}x - \frac{1}{16}x^3$$

for $-1 \leq x \leq 1$. With this in hand, we can solve $P(X < 1/2) = F_X(1/2) = \frac{99}{128}$.

3. In *most* cases where we have absolute values in probability statements, we will want to write it such that the absolute value is less than a quantity. This saves us a great deal of work! The question asks us to solve $P(|X| > 0.8)$, which we can write as $1 - P(|X| \leq 0.8) = 1 - P(-0.8 < X < 0.8)$. We can solve for $P(-0.8 < X < 0.8)$ by either integrating the PDF over the domain $[-0.8, 0.8]$ *or* using the CDF. I already found the CDF in the previous part, so I'll use that here:

$$P(|X| > 0.8) = 1 - P(-0.8 < X < 0.8) = 1 - [F_X(0.8) - F_X(-0.8)] = 0.164.$$

Question 3

(12pts) The proportion of the budget for a certain type of industrial company that is allotted to environmental and pollution control is coming under scrutiny. A data collection project determines that the distribution of these proportions is given by

$$f(y) = \begin{cases} 5(1-y)^4, & 0 \leq y \leq 1, \\ 0, & \text{elsewhere.} \end{cases}$$

1. (6pts) Verify that the above is a valid density function. [*Hint*: You may use a plot.]

2. (3pts) What is the probability that a company chosen at random expends less than 10% of its budget on environmental and pollution controls?

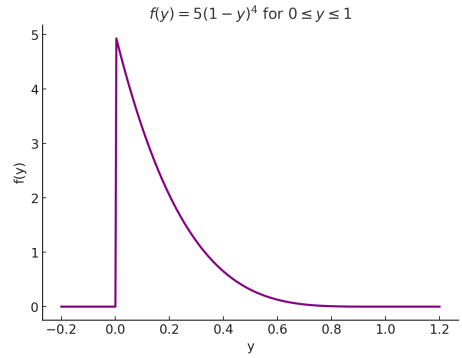
3. (3pts) What is the probability that a company selected at random spends more than half of its budget on environmental and pollution controls?

Solution 3

1. There are technically two things we need to do to ensure something is a valid PDF. The first is to ensure it integrates to 1:

$$5\int_0^1 (1-y)^4 dy = -(1-y)^5|_0^1 = 1.$$

The second thing we need to do is ensure that $f(y) \geq 0$ for all y; the easiest way to show this is with a plot.



2. We need to find $P(Y < 0.1)$. Once again, there are several ways to solve this problem: integrating the PDF over $[0, 0.1]$ or using the CDF. I'll integrate the PDF this time:

$$P(Y < 0.1) = 5\int_0^{0.1} (1-y)^4 dy = -(1-y)^5|_0^{0.1} = 0.4095.$$

3. We need to compute $P(Y > 1/2)$. Once again, let's integrate the PDF:

$$P(Y > 1/2) = 5\int_{1/2}^{1} (1-y)^4 dy = -(1-y)^5|_{1/2}^{1} = 1/32 = 0.03125.$$

Question 4

(16pts) The joint probability density function of the random variables X, Y, and Z is

$$f(x,y,z) = \begin{cases} \frac{4xyz^2}{9}, & 0 < x, y < 1, 0 < z < 3, \\ 0, & \text{elsewhere.} \end{cases}$$

Find

1. (3pts) the joint marginal density function of Y and Z;
2. (3pts) the marginal density of Y;
3. (5pts) $P(\frac{1}{4} < X < \frac{1}{2}, Y > \frac{1}{3}, 1 < Z < 2)$;
4. (5pts) $P(0 < X < \frac{1}{2} \mid Y = \frac{1}{4}, Z = 2)$.

Solution 4

1. To find the joint marginal density function of Y and Z, we need to integrate $f(x,y,z)$ over X:

$$f(y,z) = \int_0^1 \frac{4}{9} xyz^2 dx = \frac{2}{9} yz^2, \quad 0 < y < 1, \;\; 0 < z < 3.$$

2. We could integrate $f(x, y, z)$ over X and Z *or* we can integrate $f(y, z)$ over Z to obtain the marginal density of Y. The latter saves us a bit more time since we already have $f(y, z)$:

$$f(y) = \int_0^3 \frac{2}{9} y z^2 dz = 2y, \quad 0 < y < 1.$$

3. Solving this probability statement requires a direct 3-dimensional integration over the PDF, which we can separate into 3 smaller integrals:

$$\begin{aligned} P\left(\frac{1}{4} < X < \frac{1}{2}, Y > \frac{1}{3}, 1 < Z < 2\right) &= \frac{4}{9} \int_{\frac{1}{4}}^{\frac{1}{2}} \int_{\frac{1}{3}}^{1} \int_{1}^{2} xyz^2 dz dy dx \\ &= \frac{4}{9} \int_{\frac{1}{4}}^{\frac{1}{2}} x dx \int_{\frac{1}{3}}^{1} y dy \int_{1}^{2} z^2 dz \\ &= \frac{7}{162} = 0.0432 \end{aligned}$$

4. Once again, we're tasked with computing a conditional probability. If we are not given the conditional distribution, then we will need to solve for it:

$$f(x \mid y, z) = \frac{f(x, y, z)}{f(y, z)} = \frac{\frac{4}{9} x y z^2}{\frac{2}{9} y z^2} = 2x, \quad 0 < x < 1.$$

Now that we have this conditional distribution, we can solve the probability statement. Note that the conditional density for X does not actually depend on the values for Y and Z, so $P(0 < X < \frac{1}{2} \mid Y = \frac{1}{4}, Z = 2) = P(0 < X < \frac{1}{2})$, which is

$$P(0 < X < 1/2) = \int_0^{\frac{1}{2}} 2x dx = \frac{1}{4}.$$

Question 5

(10pts) A chemical system that results from a chemical reaction has two important components. The joint distribution describing the proportions X_1 and X_2 of these two components is given by

$$f(x_1, x_2) = 2, \quad 0 < x_1 < x_2 < 1.$$

1. (4pts) Give the marginal distributions of X_1 and X_2.
2. (3pts) What is the probability that component proportions produce the results $X_1 < 0.2$ and $X_2 > 0.5$?
3. (3pts) Give the conditional distribution $f_{X_1|X_2}(x_1 \mid x_2)$.

Solution 5

1. To obtain the marginal distributions, we need to integrate over the excluded random variables:

$$f(x_1) = \int_{x_1}^{1} 2dx_2 = 2(1 - x_1), \quad 0 < x_1 < 1,$$
$$f(x_2) = \int_{0}^{x_2} 2dx_1 = 2x_2, \quad 0 < x_2 < 1.$$

   Pay special attention to the bounds of integration.

2. We need to compute $P(X_1 < 0.2, X_2 > 0.5)$:

$$P(X_1 < 0.2, X_2 > 0.5) = 2\int_{\frac{1}{2}}^{1}\int_{0}^{0.2} dx_1 dx_2 = 0.2.$$

3. We are all masters at this one by now!

$$f(x_1 \mid x_2) = \frac{f(x_1, x_2)}{f(x_2)} = \frac{2}{2x_2} = \frac{1}{x_2}, \quad 0 < x_1 < x_2.$$

   A couple things to note here: (1) the conditional distribution is constant; we call this distribution a **uniform distribution** because the probability is uniform across the domain $[0, x_2]$, and (2) the support of X_1 still depends on x_2.

Question 6

(5pts) The probability distribution of the discrete random variable X is

$$f(x) = \binom{3}{x}(1/4)^x(3/4)^{3-x}, \quad x = 0, 1, 2, 3.$$

Find the mean of X.

Solution 6

For this problem, we can directly apply the definition of mathematical expectation:

$$\begin{aligned} E(X) &= \sum_{x=0}^{3} xf(x) = \sum_{x=0}^{3} x\binom{3}{x}(1/4)^x(3/4)^{3-x} \\ &= (0)\binom{3}{0}(1/4)^0(3/4)^{3-0} + \cdots + (3)\binom{3}{3}(1/4)^3(3/4)^{3-3} \\ &= 3/4. \end{aligned}$$

Question 7

(6pts) Let X and Y be random variables with joint density function

$$f(x, y) = 4xy, \quad 0 < x, y < 1.$$

Find the expected value of $Z = \sqrt{X^2 + Y^2}$.

Solution 7

This problem might be the hardest integration we've had to do thus far. We apply the definition of expectation with a function of multiple random variables, where $Z = g(X, Y)$:

$$\begin{aligned}
E[g(X,Y)] &= \int_0^1 \int_0^1 g(x,y) f(x,y) dx dy \\
&= 4 \int_0^1 y \int_0^1 x\sqrt{x^2+y^2} dx dy \\
&= 4 \int_0^1 y \left[\frac{1}{2} \int_{y^2}^{1+y^2} u^{1/2} du \right] dy \quad (\text{set } u = x^2 + y^2) \\
&= \frac{4}{3} \int_0^1 y \left[(1+y^2)^{3/2} - y^3 \right] dy \\
&= 0.9752,
\end{aligned}$$

where we use another u-substitution to solve the integral in the penultimate equality ($u = 1 + y^2$).

Question 8

(10pts) For a laboratory assignment, if the equipment is working, the density function of the observed outcome X is

$$f(x) = 2(1 - x), \quad 0 < x < 1.$$

Find the variance and standard deviation of X.

Solution 8

Recall that the convenient method for solving variance (and hence, standard deviation) is $\sigma_X^2 = E(X^2) - E(X)^2$. So, let's solve for those two quantities:

$$\begin{aligned}
E(X^2) &= \int_0^1 x^2 \cdot 2(1-x) dx = 1/6, \\
E(X) &= \int_0^1 x \cdot 2(1-x) dx = 1/3.
\end{aligned}$$

With these in hand, we can easily compute variance and standard deviation:

$$\sigma_X^2 = 1/6 - (1/3)^2 = \frac{1}{18},$$
$$\sigma_X = \sqrt{1/18} = 0.2357.$$

Question 9

(10pts) Suppose that X and Y are independent random variables having the joint probability distribution:

$f(x,y)$		x = 2	x = 4
y	1	0.10	0.15
	3	0.20	0.30
	5	0.10	0.15

Find

1. (5pts) $E(2X - 3Y)$;
2. (5pts) $E(XY)$.

Solution 9

1. Let's simplify this problem by using the result of an expectation of a linear combination of random variables: $E(2X - 3Y) = 2E(X) - 3E(Y)$. So, we just need to solve for $E(X)$ and $E(Y)$:

$$E(X) = \sum_x xf(x) = 2(0.1 + 0.2 + 0.1) + 4(0.15 + 0.30 + 0.15) = 3.2,$$
$$E(Y) = \sum_y yf(y) = 1(0.1 + 0.15) + 3(0.2 + 0.3) + 5(0.1 + 0.15) = 3.$$

   Therefore, $E(2X - 3Y) = 2(3.2) - 3(3) = -2.6$.

2. This part of the problem can be solved directly:

$$\begin{aligned} E(XY) &= \sum_x \sum_y xyf(x,y) \\ &= 1(2)(0.1) + 1(4)(0.15) + \cdots + 5(4)(0.15) \\ &= 9.6. \end{aligned}$$

Question 10

(15pts) Compute $P(\mu - 2\sigma < X < \mu + 2\sigma)$, where X has the density function

$$f(x) = 6x(1-x), \quad 0 < x < 1,$$

and compare with the result given in Chebyshev's theorem. Round decimals to four places.

Solution 10

In order to solve for $P(\mu - 2\sigma < X < \mu + 2\sigma)$, we need to know $\mu = E(X)$ and $\sigma^2 = E(X^2) - E(X)^2$, so let's compute the necessary quantities:

$$E(X) = 6\int_0^1 x^2(1-x)dx = 1/2,$$
$$E(X^2) = 6\int_0^1 x^3(1-x)dx = 0.3.$$

Therefore, $\mu = 1/2$ and $\sigma = \sqrt{0.3 - (1/2)^2} = \frac{1}{\sqrt{20}}$. Now, we solve

$$\begin{aligned} P(\mu - 2\sigma < X < \mu + 2\sigma) &= P(1/2 - 2/\sqrt{20} < X < 1/2 + 2/\sqrt{20}) \\ &= \int_{1/2-2/\sqrt{20}}^{1/2+2/\sqrt{20}} 6x(1-x)dx \\ &= 0.9839. \end{aligned}$$

Chebyshev's theorem states that the lower bound of the probability of X being within 2 standard deviations is $1 - 1/2^2 = 0.75$. Clearly, our computation satisfies that lower bound. In fact, because we have a probability distribution, we are able to compute the probability explicitly. While Chebyshev's theorem would tell us that it's at least 75%, our computation allows us to confidently claim it's 98.39%.

Question 11

(10pts) Consider the joint density function

$$f(x,y) = \frac{16y}{x^3}, \quad x > 2, \;\; 0 < y < 1.$$

Compute the covariance σ_{XY}.

Solution 11

We want to compute $\sigma_{XY} = E(XY) - E(X)E(Y)$. We see that

$$E(XY) = 16 \int_2^\infty x^{-2} dx \int_0^1 y^2 dy = 8/3.$$

To find $E(X)$ and $E(Y)$, we'll need to obtain the marginal densities:

$$f(x) = 16x^{-3} \int_0^1 y dy = 8x^{-3}, \quad x > 2,$$
$$f(y) = 16y \int_2^\infty x^{-3} dx = 2y, \quad 0 < y < 1.$$

Now, we can compute the marginal expected values:

$$E(X) = 8 \int_2^\infty x^{-2} dx = 4,$$
$$E(Y) = 2 \int_0^1 y^2 dy = 2/3.$$

Finally, we see that $\sigma_{XY} = 8/3 - 4(2/3) = 0$.